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Towards quantification of drugs in exhaled breath aerosol

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1 Introduction

Human breath contains not only inorganic gases and water vapor, but also a variety of organic compounds. Some of these compounds are transferred from the blood to the breath in the alveoli, providing valuable information about the body's health.

In addition to volatile organics, non-volatile compounds are also present in breath at even lower concentrations and are believed to be exhaled in aerosol form. Measuring these compounds, therefore, requires preconcentration combined with the most sensitive techniques available to detect them in the part per trillion (ppt) range in breath. The following concentrations need to be detected:

Table 1: Concentrations of nonvolatile compounds in breath [1,2]

Aerosol particles	Lipids (mainly DPPC)	Drugs
400 - 2100 pg / L breath	40 - 210 pg / L breath	fg – pg / L breath

2 Method Overview

As part of a randomized controlled trial at the Institute of Forensic Medicine in Zürich to determine the driving performance under the influence of attention-deficit/hyperactivity disorder (ADHD) drugs, we are going to measure a variety of drugs to treat ADHD in breath, and correlate these measurements with blood concentrations.

Pharmacokinetic pre-study

As proof of concept, we preformed a preliminary study in which we monitored a volunteer who takes Lisdexamfetamine (Elvanse) every morning. In this study, we measured the volunteer's blood and breath at several timepoints throughout the day.

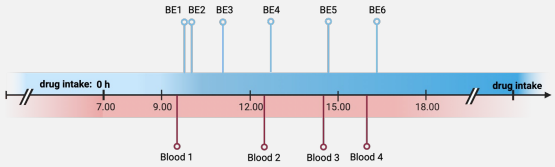


Figure 1: Timeline of pre-study, with labeled blood and breath samples (BE)

Lisdexamfetamine is metabolized in the blood to form the active drug amphetamine (amp), which was analyzed in the blood using a routine LC-MS method (with a calibration range of 2.4 to 800 ng/mL).

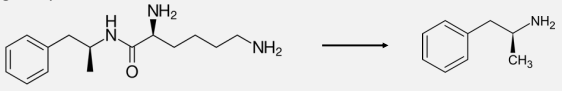


Figure 2: Metabolization of lisdexamfetamine into (dextro)amphetamine

Breath measurements and analysis

Breath samples were taken with the collection device Breath Explor (each time 12 deep exhalations). Afterwards they were stored at -18°C until analysis. One of the three cartridges was extracted with 1.9 mL of methanol (including internal standard and ethylene glycol). The sample was up-concentrated in a vacuum centrifuge and dissolved in 90 µL methanol : water (1:1) mixture and analyzed using liquid chromatography coupled to a mass spectrometer (LC-MS).

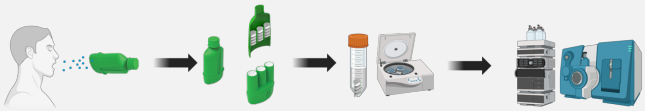


Figure 3: Workflow from sampling to analysis. Breath is sampled with Breath Explor and stored at -18°C. One of the three cartridges is extracted, up-concentrated and analyzed using an optimized MRM method on a QTRAP (QTRAP 5500 - SCIEX)

Table 2: Specification LC-MS method

Mobile phase A	H_20, 5% MeOH, 4 mM NH_4FA, 0.1% NH_3	Mobile phase B	MeOH, 5% H_20, 4 mM NH_4FA, NH_3
Injection volume	10 µL	Gradient:	5% B to 99% B (total 22 min)
Column	C18 1.7µm 2.1 x 50 mm	MRM transition	136.3 m/z → 119.1 m/z

3 Results and Discussion

Pharmacokinetic behavior / Blood-Breath correlation

A clear trend showing the elimination of amphetamine over time can be seen in both breath and blood concentrations. The trend is similar to the expected behavior (simulated pharmacokinetic shown in figure 5).

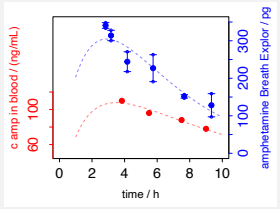


Figure 4: Measured pharmacokinetic in blood (left axis) and breath (right axis).

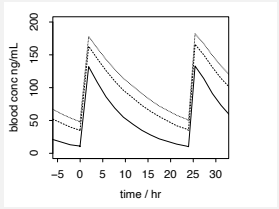


Figure 5: Simulated pharmacokinetic (f: 0.8, t_{max}: 1.5 h, V_d: 5.6 L/kg, $t_{1/2}$ 6/10/12h)

Table 3: Measured and calculated values of method

LOD_{Amp}	xx
blood c_{Amp}	78 – 110 ng / mL
breath c_{Amp}	97 – 348 pg / filter 6 - 21 pg / L breath 0.3 -1.2 ppt

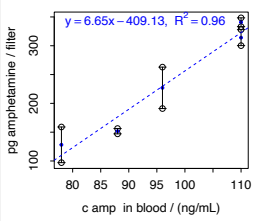


Figure 6: Blood–breath correlation

4 Conclusion

- A correlation between the concentration of amphetamine in the blood and the breath samples (from one person) was observed.
- Further optimization of the measurement method to quantify endogenous lipids (e.g. dipalmitoyl phosphatidylcholine DPPC) is required.

References

[1] P. Larsson et al. Respiratory Physiology & Neurobiology 243 (2017) 39–46

[2] H. Sinapour et al. Journal of Chromatography B 1240 (2024) 124142

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